THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. On a map, 1 cm represents 20 km. A distance of 6.5 cm on the map represents: 1

(a) 260 km **(b)** 130 km **(c)** 150 km **(d)** 65 km

2. On a map, 1 cm represents 40 km. A distance of 5 cm on the map represents: 1

(a) 100 km **(b)** 240 km **(c)** 200 km **(d)** 400 km

3. ₹ 1500 is shared among A, B and C in the ratio 5:3:2. B gets: 1

(a) ₹ 750 **(b)** ₹ 450 **(c)** ₹ 300 **(d)** ₹ 500

4. 6 workers finish a job in 20 days. How many days will 15 workers take? 1

(a) 50 **(b)** 8 **(c)** 7 **(d)** 10

5. 9 workers finish a job in 16 days. How many days will 12 workers take? 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 21.3333

6. In a pie chart, a sector representing 15% of the data has a central angle of: 1

(a) 54° **(b)** 15° **(c)** 64° **(d)** 30°

7. If the speed doubles, the time taken for the same distance: 1

(a) halves **(b)** doubles **(c)** triples **(d)** remains same

8. A whole divided in the ratio 3 : 7 gives the smaller part as what fraction of the whole? 1

(a) 7/10 **(b)** 3/4 **(c)** 3/7 **(d)** 3/10

9. If the ratio of boys to girls is 3 : 2, the percentage of girls is: 1

(a) 40% **(b)** 60% **(c)** 33.3% **(d)** 50%

10. If x varies inversely with y and x = 6 when y = 4, then when y = 8, x = 1

(a) 24 **(b)** 12 **(c)** 3 **(d)** 2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. 6 men build a wall in 12 days. How long will 9 men take? 2

12. A pie chart sector is 72°. What percent of the whole is it? 2

13. Express the scale 1 : 200000 as 1 cm = ? km. 2

14. If the ratio of red, white and blue balls is 2 : 3 : 5 and there are 40 balls, how many are blue? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. On a map, 1 cm stands for 40 km. Two cities are 7.5 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

16. On a map, 1 cm stands for 15 km. Two cities are 9 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

17. On a map, 1 cm stands for 25 km. Two cities are 12.4 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

OR

On a map, 1 cm stands for 150 km. Two cities are 3.2 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

18. On a map, 1 cm stands for 50 km. Two cities are 6.8 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

OR

On a map, 1 cm stands for 30 km. Two cities are 11 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A pie chart shows the time spent by a student in a 24-hour day: sleep 8 h, school 6 h, study 4 h, play 3 h, other 3 h. 5

(i) Find each angle.

(ii) Which sector is the largest?

(iii) By how many degrees does study exceed play?

OR

(i) A hostel has food for 80 students for 30 days. After 10 days 20 students leave. For how many more days will the food last? 5

(ii) Explain.

20. (i) Three friends invested ₹ 8000, ₹ 12000 and ₹ 20000. Profit is ₹ 10000. Find each share. 5

(ii) What if the first invests for 12 months, others for 6 months?

OR

Case study: A model of a building is made at a scale of 1 : 150. 5

(i) A 45 m tall building is how tall on the model?

(ii) The model's door is 1.6 cm; find the real door.

(iii) Floor area ratio between model and real.

END OF QUESTION PAPER — Total: 40 Marks